bigquery-public-data.new_york_citibike

Two tables in a simple star: one large trip fact table and one station dimension. They were collected years apart from different Citi Bike systems, which shapes every join and KPI you build from them.

FACT · HISTORICAL · STATIC

citibike_trips

Every Citi Bike ride: when, where from and to, which bike, and who rode.

Grain	One row per trip (no trip ID)
Rows	58,937,715 total · **53,108,721 real** · 5,828,994 blank
Columns	16 (timing 3 · station 8 · bike 1 · rider 4)
Time span	2013-07-01 to 2018-05-31 (NYC local)
Size	8.0 GB logical · no partitioning
Last updated	September 2018
Station key	`start_station_id`, `end_station_id` (INT, legacy)

DIMENSION · LIVE SNAPSHOT

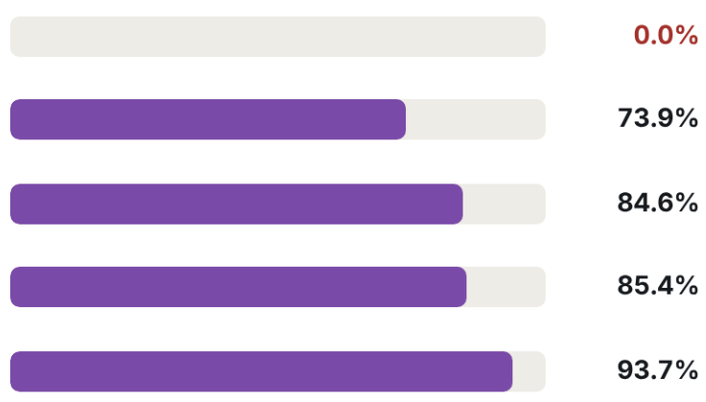
citibike_stations

Each station's name, location, dock capacity and current status.

Grain	One row per station (current state only)
Rows	2,518
Columns	18 (identity/location 9 · live status 9)
Time span	Status as of last refresh (Jul to Nov 2026)
Size	0.4 MB
Last updated	2026-11-03 (refreshed continuously)
Station key	`station_id` (STRING, UUID or long numeric)

Share of trips that can be linked to a station record

By join method, out of 53.1M real trips (start station)



What to know before you query

- **IDs don't line up.** Trips use integer IDs (72 to 3686); stations use UUID strings. Join through a name + location crosswalk (see ERD).
- **10% of trip rows are blank.** Filter `starttime IS NOT NULL` first.
- **Different eras.** Station status is from 2026; trips end in 2018. Use stations for location and capacity, not past availability.
- **Dead columns:** `customer_plan` (always blank), both `eightd_*` flags (always false), `rental_methods` (constant).
- **Cost:** trips is 8 GB with no partitions. Select only needed columns; a 4-column profile query scanned 1.6 GB.

TABLE DOCUMENTATION · BIGQUERY-PUBLIC-DATA.NEW_YORK_CITIBIKE

citibike_trips

Fact table. One row per completed Citi Bike ride in New York City, July 2013 through May 2018. Station details are copied (denormalized) onto every trip, and there is no trip ID column. Static: last modified September 2018.

58.94M	53.11M	5.83M	881 / 926	17,682
total rows (8.0 GB logical)	real trips after removing blank rows	fully blank rows (9.9%)	distinct start / end station IDs	distinct bikes

#	COLUMN	TYPE	KEY	NULL / BLANK	PROFILE (53.1M REAL TRIPS)
TRIP TIMING					
1	**tripduration**	INTEGER		9.9%	Seconds. Min 60, P25 383, **median 628 (10.5 min)**, P75 1,062. Max 19.5M s (225 days); **11,513 trips over 24 h** are outliers to cap or drop.
2	**starttime**	DATETIME		9.9%	NYC local time, no time zone. 2013-07-01 to 2018-05-31. Trips per year: 5.0M (2013, half yr), 8.1M, 9.9M, 10.3M, 14.1M, 5.7M (2018, 5 mo).
3	**stoptime**	DATETIME		9.9%	NYC local time. 1.62M rows where stoptime minus starttime differs from tripduration by more than 2 s; use `tripduration` as the source of truth.
START STATION (DENORMALIZED COPY OF STATION ATTRIBUTES)					
4	**start_station_id**	INTEGER	FK*	9.9%	Legacy integer ID, range 72 to 3686. **Does not match citibike_stations.station_id** (see ERD); join through a station crosswalk.
5	**start_station_name**	STRING		9.9%	918 distinct names for 881 IDs: 47 stations were renamed during 2013 to 2018.
6	**start_station_latitude**	FLOAT		9.9%	WGS 84 decimal degrees. 41 real trips have a 0 or missing coordinate. Used for the geographic fallback join.
7	**start_station_longitude**	FLOAT		9.9%	
END STATION (SAME STRUCTURE AS START)					
8	**end_station_id**	INTEGER	FK*	9.9%	926 distinct IDs (45 more than start). **1.14M round trips** where end = start station.
9	**end_station_name**	STRING		9.9%	Same naming caveats as start_station_name.
10	**end_station_latitude**	FLOAT		9.9%	WGS 84 decimal degrees.
11	**end_station_longitude**	FLOAT		9.9%	
BIKE AND RIDER					
12	**bikeid**	INTEGER	ID	9.9%	17,682 distinct bikes. No bike dimension table exists; usable for fleet utilization (trips or hours per bike).
13	**usertype**	STRING		9.9%	**Subscriber 46.9M (88.3%)** annual members · Customer 6.2M (11.7%) 24-hour or 7-day pass.
14	**birth_year**	INTEGER		19.3%	Range 1874 to 2002. 22,871 trips list a year before 1920 (implausible, likely default entries). Also null for 5.6M real trips.
15	**gender**	STRING		9.9%	male 35.6M (67.1%) · female 11.4M (21.4%) · unknown 6.1M (11.5%).
16	**customer_plan**	STRING		100%	**Empty in every row.** Exclude from analysis.

Grain and keys

- **Grain:** one row per trip. No primary key column; if you need one, build it with `FARM_FINGERPRINT` over bikeid + starttime + start_station_id.
- **FK*** = logical foreign key only. BigQuery does not enforce it, and the IDs need a crosswalk to reach citibike_stations.
- **Partitioning / clustering:** none. Every query scans the full column, so select only the columns you need (about 0.5 to 1 GB per column).

Standard cleaning filter

```
WHERE starttime IS NOT NULL          -- drops 5.83M blank rows
  AND tripduration BETWEEN 60 AND 86400  -- drops 11,513 trips over 24 h
-- ignore customer_plan (always blank)
-- treat birth_year < 1920 as NULL
```

TABLE DOCUMENTATION · BIGQUERY-PUBLIC-DATA.NEW_YORK_CITIBIKE

citibike_stations

Dimension table (live snapshot). One row per Citi Bike station with its location, capacity and current status, taken from the GBFS station feed. Google refreshes it continuously: it was last modified 2026-11-03, so the status columns describe *today*, not the 2013 to 2018 trip period.

2,518	74,497	31,486	2,463	48
stations (rows)	total docks (sum of capacity)	bikes available at snapshot	installed (97.8%)	never reported (1970 timestamp)

#	COLUMN	TYPE	KEY	NULL	PROFILE (2,518 ROWS)
IDENTITY AND LOCATION (STATIC ATTRIBUTES)					
1	**station_id**	STRING	PK	0%	REQUIRED and unique. Two formats: **1,809 UUIDs** (66dc292c-0aca-11e7-...) and **709 long numeric strings** (1986707977001495180). Neither matches the integer IDs in citibike_trips.
2	**name**	STRING		0%	Public name, e.g. "Broadway & W 41 St". 2,517 distinct (one name is shared by 2 stations). Best text key for matching to trips.
3	**short_name**	STRING	AK	0%	Unique station code, e.g. "6560.01". Alternate key; not present in trips.
4	**latitude**	FLOAT		0%	WGS 84 decimal degrees. Used for the geographic fallback join (nearest station within 50 to 100 m).
5	**longitude**	FLOAT		0%	
6	**region_id**	INTEGER		0%	4 values: **71 (2,400 stations)**, 70 (78), 311 (28), 0 (12). No region lookup table in the dataset.
7	**rental_methods**	STRING		0%	Constant "KEY, CREDITCARD". Stored as a string even though the description says array. No analytic value.
8	**capacity**	INTEGER		0%	Total docks, max 123. **38 stations show 0**; treat as unknown, not as empty.
9	**eightd_has_key_dispenser**	BOOLEAN		0%	Always false. No description. Exclude.
CURRENT STATUS (CHANGES EVERY REFRESH)					
10	**num_bikes_available**	INTEGER		0%	Bikes ready to rent now. Network total 31,486.
11	**num_bikes_disabled**	INTEGER		0%	Broken or held bikes docked now. Network total 7,950.
12	**num_docks_available**	INTEGER		0%	Open docks accepting returns. Network total 32,840.
13	**num_docks_disabled**	INTEGER		0%	Empty but out-of-service docks. Network total 460.
14	**is_installed**	BOOLEAN		0%	On the street: 2,463 true, 55 false.
15	**is_renting**	BOOLEAN		0%	Renting bikes: 2,431 true.
16	**is_returning**	BOOLEAN		0%	Accepting returns.
17	**eightd_has_available_keys**	BOOLEAN		0%	Always false. No description. Exclude.
18	**last_reported**	TIMESTAMP		0%	Last status report. Real values run 2026-07-16 to 2026-11-03; **48 rows hold 1970-01-02**, a placeholder meaning "never reported". Filter these before using status columns.

Grain and keys

- **Grain:** one row per station, current state only. No history is kept, so a station's capacity or status in 2015 is not recoverable here.
- **PK** station_id (unique, required). **AK** short_name is also unique.
- **Coverage gap:** 2,518 stations today versus 926 in the trip history. Today's network includes later expansions (for example Bronx stations like Southern Blvd & E 149 St) that trips can never reference.

Use it for

- **Station attributes** on trips: capacity, region, coordinates (via the crosswalk).
- **Current-state KPIs:** fleet availability, % bikes disabled (7,950 / 39,436 docked = 20.2%), stations empty or full right now.
- **Not for:** historical status or capacity during 2013 to 2018.